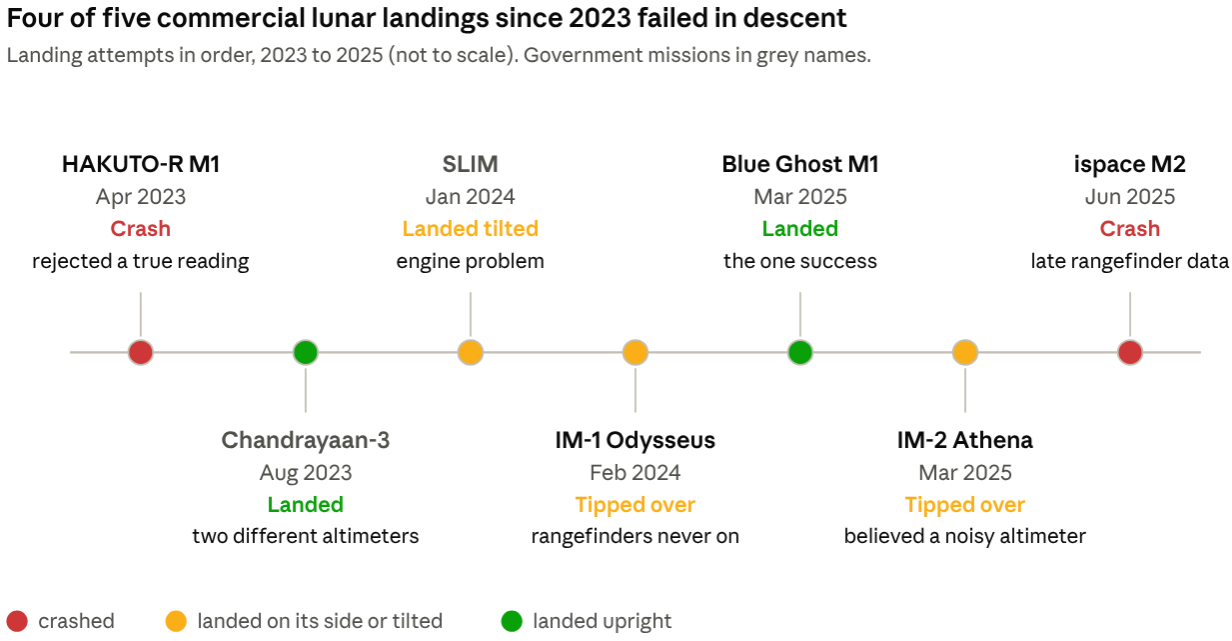
# PRJ1 Literature Review: Sensor Trust for Lunar South Pole Landing

Four of the five commercial lunar landings attempted between 2023 and 2025 crashed or tipped over. In all four, the altitude or terrain data used in terminal descent was wrong, missing or rejected; Firefly's Blue Ghost M1 (March 2025) was the one success. This review covers seven themes: mission failures, NASA priorities, descent navigation architectures, South Pole terrain and illumination data, measurement rejection in Kalman filters, the closest prior work, and a targeted search of AIAA SciTech, IEEE Aerospace and ION papers. Section 8 summarises every source in one line each.

## 1. Mission evidence: navigation failures in terminal descent




lunar landing attempts 2023–2025 · 7 missions

Every commercial failure involved altitude data: rejected, missing, noisy or late. The table below gives the reported causes.

| Mission | Date | Outcome | Navigation cause (as reported) | Source |
|---|---|---|---|---|
| ispace M2 *RESILIENCE* | 6 Jun 2025 | Crash, Mare Frigoris | Laser rangefinder was slow to produce valid readings, so the lander could not decelerate in time; candidate causes include low surface reflectivity, in-flight degradation and high descent speed | SpacePolicyOnline, Payload |
| IM-2 *Athena* (Intuitive Machines) | Landed ~6 Mar 2025 | Landed on its side near the South Pole; ~12 h of operations | Laser altimeter signal noise and distortion; long shadows from low Sun; LRO crater imagery from 100 km did not match crater appearance at low altitude | SpacePolicyOnline, Spaceflight Now |
| IM-1 *Odysseus* (Intuitive Machines) | 22 Feb 2024 | Broke a leg and tipped over near Malapert A (about 80°S); operated ~6 days | Laser rangefinders unusable because a ground safety switch was not flipped before launch; team patched in NASA's Navigation Doppler Lidar (NDL) experiment in lunar orbit | New Atlas, The Space Review |

| Mission | Date | Outcome | Navigation cause (as reported) | Source |
|---|---|---|---|---|
| HAKUTO-R M1 (ispace) | 25 Apr 2023 | Crash, Atlas crater | Altimeter reading jumped when crossing a ~3 km crater-rim cliff; onboard filter rejected it as a sensor fault; lander believed it was at 0 km while ~5 km up, descended until propellant ran out, then free-fell | ispace debriefing, Gizmodo |

HAKUTO-R is the clearest case for this project. Its altitude-rejection filter was a deliberate "robust measure" against hardware faults, and ispace said its simulations did not model the terrain on the route after a change of landing site. IM-1, IM-2 and ispace M2 show the second failure mode: the lander could not judge which sensor to believe when one degraded.

The 2025 failures show the mirror image. On IM-2 the rangefinder picked up noise even with its laser off; the software believed the lander was still about 400 m up, and it hit the ground at 10–13 m/s (Aerospace America, 22 Jul 2025). **In 2023 the software rejected a correct sensor; in 2025 it believed a faulty one.** Both are failures of judgement, and the trust layer addresses both.

HAKUTO-R's site, Atlas crater, is in the northern hemisphere, not the South Pole. The South Pole evidence comes from IM-1 (near Malapert A, about 80°S) and IM-2, which landed closer to the pole than any spacecraft before it.

Two government missions show the alternatives. ISRO's Chandrayaan-3 (Aug 2023) flew both a laser and a Ka-band microwave altimeter; its designer says the microwave altimeter's wider footprint helped it tolerate abrupt altitude changes (Scientific American). JAXA's SLIM (Jan 2024) matched craters against a pre-built map from Kaguya images, and all 14 image-matching fixes were judged accurate (JAXA briefing).

## 2. NASA technology priorities

NASA STMD's FY26 Civil Space Shortfall Prioritization (2026) consolidates technology needs into 32 shortfalls and selects 40 need statements as STMD's FY26 focus areas. Four of those 40 match this project:

- **3.02** Land on lunar south pole exploration sites in various illumination conditions with appropriate accuracy (references ESDMD #1101, landing accuracy). Listed first among the 40.
- **3.03** Perform stable touchdown of large vehicles on uneven, sloped terrain.
- **24.03** Autonomous orbit determination, navigation and pointing.
- **24.05** A lunar position, navigation and timing (PNT) architecture.

In the stakeholder ranking, the parent shortfalls rank **SF03** (precise lunar landing) 15th and **SF24** (tracking and navigation) 8th of 32.

## 3. Descent navigation architectures

- **NASA SPLICE** (Safe and Precise Landing – Integrated Capabilities Evolution), started FY2018, combines terrain-relative navigation (TRN), Navigation Doppler Lidar and a hazard-detection lidar, and was flight-tested on Blue Origin New Shepard (SPLICE poster). Its target is landing within 50 m of the intended site.
- **Nova-C hazard detection** (Getchius et al., 2022): the lander must confirm a site free of hazards over 30 cm with slope under 10°, running hazard detection about 400 m above the target within 15 s (paper). This is the lander family that tipped over on IM-1 and IM-2.
- **Lidar TRN**: NASA field tests show lidar-based TRN delivering position estimates under 90 m and less sensitivity to lighting than camera TRN (CORE record).

**Onboard terrain mapping is emerging.** SPLICE's hazard-detection lidar maps the ground near the landing site during the final descent, and Adams et al. (2025) note that improving lidar now lets landers generate terrain maps onboard. A live scan matched against the stored map can correct the lander's position and spot hazards the stored map misses. This motivates our optional Extension 1 (live 3D scan).

## 4. South Pole terrain and illumination data

- **Barker et al. (2021)**, *Planetary and Space Science* 203: 5 m/pixel LOLA elevation maps of South Pole landing sites, with orbit errors cut by more than 10x and per-pixel height and slope uncertainty maps (NTRS). Released as PGDA Product 78.
- **Mazarico et al. (2011)**, *Icarus*: illumination of the lunar polar regions from 240 m LOLA maps over an 18.6-year cycle at hourly steps (NTRS). Basis of PGDA Product 69.
- **Bertone et al.**, *The Planetary Science Journal*: Shape-from-Shading 5 m/pixel DEMs that sharpen LOLA maps with LROC NAC images over 4,500+ km² of the 13 original Artemis III candidate regions, with slope, roughness and maximum-illumination products (PGDA Product 104; data released 16 Dec 2025 on Zenodo).

These datasets mean a lander can know, before launch, where cliffs and shadows lie along its path, with an error estimate for every pixel.

## 5. Measurement rejection and sensor degradation

- **Innovation gating**: a standard Kalman filter rejects a measurement when its normalised innovation exceeds a fixed chi-square threshold (Bar-Shalom, Li and Kirubarajan, 2001). The threshold assumes the terrain model is accurate between updates, which a 3 km cliff breaks.
- **Terrain-aided navigation (SITAN)**: Hostetler and Andreas (1983) linearise the DEM slope under the vehicle so an altimeter can correct position. Built for aircraft over smooth terrain; the linearisation breaks at cliffs.
- **Adaptive filtering**: Mehra (1972) adjusts noise covariances from the data stream itself. It reacts after the fact rather than using a map to anticipate change.
- **Plume-dust degradation**: the University of Michigan ARC-LIGHT team (8 students, team lead A. Cushen; winner of NASA's 2024 Human Lander Challenge) built a 0.91 x 0.91 x 1.2 m chamber with a downward camera and lidar, filled it with a fine mist standing in for plume dust, and trained a small neural network to estimate dust thickness from camera images. That estimate corrects the lidar's distances, and the system shuts itself off when its error grows too large (U-M CLaSP, *Aerospace* 2025). Sensor trust should change with descent phase, not only with hardware health.

## 6. Closest prior work

The building blocks are published and recent, but we found no paper that sets a lander filter's outlier gate or sensor weights ahead of time from terrain and illumination maps. Newest first:

| Work | Year | What it does | Overlap with our idea | What it does not do |
|---|---|---|---|---|
| Prakash D. and Raha, co-state data fusion and risk-aware filtering | Apr 2026, preprint | Detects EKF model inconsistency on real lunar powered-descent telemetry, earlier than standard statistical alarms | Separating "filter wrong" from "sensor wrong" | No terrain map or lighting; reactive |

| Work | Year | What it does | Overlap with our idea | What it does not do |
|---|---|---|---|---|
| Sheppard, Reale and Skinner, stochastic terrain maps for safe lunar landing | Dec 2025, IEEE Aerospace 2026 | Gaussian-process elevation maps from LRO DEMs, with noise learned from DEM confidence maps | Uses DEM uncertainty as a prior | For hazard detection and site selection, not filter trust |
| Karlgaard et al., NDL performance and IM-1 trajectory reconstruction | 2025, AAS/AIAA ASC and NASA TM | Reconstructs IM-1's last 3 minutes with NDL + IMU in an Unscented Kalman Filter, a lunar DEM and lidar beam ray-tracing; agreement within about 5 m and 0.5 m/s | Proves a DEM-based altimeter model works on flight data | Post-flight and offline |
| Cowan et al., vision-guided optic flow navigation for small lunar missions | Nov 2025, preprint | Optical flow plus laser rangefinder over South Pole terrain | Camera and rangefinder fusion | No terrain jumps or shadow handling |
| ispace, HAKUTO-R M1 debriefing | 2023 | Flight timeline: rangefinder updates from about 15 km; a gap of over 3 km "not reflected" at 16:38 UTC; estimated 0 km vs measured about 5 km at 16:43 | Primary evidence of the failure | Fixes are software tuning and wider terrain simulation |

# 7. Conference search: AIAA SciTech, IEEE Aerospace, ION (2022–2026)

The closest conference paper is NASA/Draper's SciTech 2025 work for SPLICE. It already puts a South Pole DEM inside the altimeter model, but flags faults with a **fixed** 99% chi-square gate.

| Work | Venue | Gating or weighting | Terrain or lighting used? | Gap left open |
|---|---|---|---|---|
| Adams, Ward, Smith, Pedrotty, Mendeck, Topographic slant range modeling and fault detection | AIAA SciTech 2025 (NASA JSC + Draper) | PITFAL: chi-square test on each beam's apparent altitude, fixed limit 6.635 (99%, 1 degree of freedom); a flagged beam's noise is inflated until it passes | DEM-based slant-range model; 1,000 Monte Carlo runs on a 10 x 10 km patch of the PGDA NPA South Pole DEM at 5 m; PITFAL activates on high-gradient terrain | Fixed gate, reactive response; needs a multi-beam sensor; no lighting or camera trust |
| Tomasicchio, Del Prete, Renga, Grassi, Ceresoli, Silvestrini, Lavagna et al., PVT estimation for safe lunar landing | ION GNSS+ 2024 | Preliminary Mahalanobis-distance check | Altimeter noise fixed at 1% of height | Fixed gate; no terrain or lighting |
| Shoemaker et al., TRN using autoNGC | AIAA SciTech 2022 (NASA GSFC) | Standard EKF updates | Times TRN images so local Sun elevation is about 20° to 40° | Lighting handled by planning only; at the South Pole (1.5° to 3°) that window does not exist |

| Work | Venue | Gating or weighting | Terrain or lighting used? | Gap left open |
|---|---|---|---|---|
| Cauligi et al. (JPL), ShadowNav | IEEE Aerospace 2023 | Crater-edge match scoring | External illuminator and stereo camera to see craters in shadow | Hardware fix for rovers |
| Candan and Servadio, deep learning crater TRN | arXiv, Jun 2026 | Consensus outlier removal; fixed pixel noise for craters | None | Camera noise does not change with lighting |

**Search note:** searches for "adaptive gating", "adaptive chi-square" and "illumination-dependent measurement covariance" in lunar landing found no matching paper. IEEE Xplore and AIAA ARC proceedings are paywalled; check them through the college library before claiming novelty.

## 8. Summary of each source

One line per source on what it did, what it found, and how this project uses it. Numbers match the reference list below.

### Mission reports and news

| # | Source | What it did or reported | Key finding | How we use it |
|---|---|---|---|---|
| 1 | ispace, HAKUTO-R M1 debriefing (2023) | Flight timeline of the April 2023 crash | Software rejected a correct altimeter reading at a ~3 km crater rim as a "sensor malfunction"; terrain on the new route was not in pre-flight simulations | Core case for cliff-aware gating and automatic re-testing after a site change |

| # | Source | What it did or reported | Key finding | How we use it |
|---|---|---|---|---|
| 2 | SpacePolicyOnline, ispace Mission 2 (2025) | ispace's cause analysis of the June 2025 crash | Laser rangefinder was slow to give valid readings, so the lander could not slow down in time | Delayed-reading fault scenario |
| 3 | SpacePolicyOnline and Spaceflight Now, IM-2 (2025) | Intuitive Machines' post-mission review | Altimeter noise, long low-Sun shadows and mismatched crater images caused the tip-over | Shadow-aware camera trust and noisy-altimeter scenario |
| 4 | New Atlas and The Space Review, IM-1 (2024) | Account of the Odysseus landing | Rangefinders never switched on (safety switch); NASA's NDL patched in; lander tipped over | Limit of software: dropout scenario |
| 5 | Scientific American, Chandrayaan-3 (2023) | Interview with ISRO's altimeter designer | A second, wide-footprint microwave altimeter helped tolerate abrupt altitude changes | Hardware alternative to compare against |
| 6 | JAXA SLIM briefing (2024) | Results of Japan's pinpoint landing | Crater matching against a pre-built Kaguya map; all 14 fixes judged accurate | Evidence that onboard maps work |
| 7 | Freeman (2025), *J. Space Operations & Communicator* | Qualitative review of 12 news reports on robotic landers | Leg stability most cited; altimeter or rangefinder issues in 5 of 12 | Background only |

| # | Source | What it did or reported | Key finding | How we use it |
|---|---|---|---|---|
| 29 | Button (2025), *Aerospace America* (AIAA) | Interview on IM-2 causes and IM-3 fixes | Rangefinder picked up noise even with the laser off; software believed it was ~400 m up and the lander hit at 10–13 m/s; IM-3 adds rangefinders from two vendors and more calibration orbits | Proof that the software's belief, not only the sensor, failed |

## NASA priorities and architectures

| # | Source | What it did | Key finding | How we use it |
|---|---|---|---|---|
| 8 | NASA STMD FY26 Shortfall Prioritization (2026) | Ranks 32 technology shortfalls and picks 40 focus areas | Need 3.02 (South Pole landing in any lighting) listed first of 40; SF11 ranked 4th, SF03 15th | Shows the problem is an official NASA priority |
| 9 | SPLICE project poster (NASA) | Describes NASA's precision-landing system | Combines camera TRN, Navigation Doppler Lidar and hazard-detection lidar; targets landing within 50 m | State of the art we position against |
| 10 | Getchius et al. (2022) | Nova-C hazard detection design | Clears hazards over 30 cm and slopes over 10°, scanning ~400 m above the site in 15 s | Source of our 10° tilt and slope limits |
| 11 | NASA lidar TRN study | Field tests of lidar-based terrain navigation | Position estimates under 90 m, less sensitive to lighting | Lidar vs camera trade-off |

## Terrain and illumination data

| # | Source | What it did | Key finding | How we use it |
|---|---|---|---|---|
| 12 | Barker et al. (2021), *Planetary and Space Science* | Re-processed LOLA laser data for South Pole sites | 5 m/pixel maps; orbit errors cut more than 10x; per-pixel height and slope error maps | Our terrain and the map-error term |
| 13 | PGDA Product 78 (Barker) | Publishes those maps as GeoTIFFs | Site04 Shackleton rim, Site01 connecting ridge, plus error maps | Our dataset (downloaded and checked) |
| 14 | Mazarico et al. (2011), *Icarus* | Computed polar sunlight from 240 m LOLA maps over 18.6 years | Where and how often the poles are lit or shadowed | Basis of our shadow method |
| 15 | Bertone et al., *Planetary Science Journal* (PGDA 104) | Sharpened LOLA maps with Shape-from-Shading from LROC images | 5 m maps of 13 Artemis III regions, 4,500+ km², released Dec 2025 | Optional higher-detail terrain |

## Filtering theory

| # | Source | What it did | Key finding | How we use it |
|---|---|---|---|---|
| 16 | Bar-Shalom, Li and Kirubarajan (2001) | Standard estimation textbook | Chi-square innovation gating with a fixed threshold | The baseline rule we improve on |
| 17 | Mehra (1972), *IEEE TAC* | Adaptive Kalman filtering | Re-estimates noise from the data stream | Reactive alternative we contrast with |
| 18 | Hostetler and Andreas (1983), *IEEE TAC* | SITAN terrain-aided navigation | Uses local terrain slope so an altimeter can correct position | Classic method; slope approximation breaks at cliffs |

## Closest prior work

| # | Source | What it did | Key finding | How we use it |
|---|---|---|---|---|
| 19 | Adams et al. (2025), AIAA SciTech (PITFAL) | DEM-based altimeter model plus fault check for SPLICE | Fixed 99% chi-square gate; flagged readings get inflated noise; 1,000 Monte Carlo runs on a PGDA South Pole DEM | Main comparison baseline |
| 20 | Karlgaard et al. (2025), NASA | Reconstructed IM-1's last 3 minutes | NDL + IMU in an Unscented Kalman Filter with a DEM and lidar ray-tracing; within ~5 m and 0.5 m/s | Validates DEM-based altimeter models |
| 21 | Prakash D. and Raha (2026), arXiv | New consistency monitor on real lunar descent data | Detects filter inconsistency earlier than standard alarms | Alternative fault detection |
| 22 | Sheppard et al. (2025), IEEE Aerospace 2026 | Gaussian-process terrain maps from LRO DEMs | Learns spatially varying map noise from DEM confidence | Map-uncertainty modelling idea |
| 23 | Cowan et al. (2025), arXiv | Optical flow plus laser rangefinder for small landers | Works over South Pole terrain with simple terrain models | Low-cost sensor suite |
| 24 | Tomasicchio et al. (2024), ION GNSS+ | Camera, IMU, altimeter and lunar navigation satellites | Mahalanobis outlier check; altimeter noise fixed at 1% of height; errors of tens of metres | Shows fixed gates are standard practice |
| 25 | Shoemaker et al. (2022), AIAA SciTech | Terrain-relative navigation in NASA's autoNGC software | Times camera fixes for a 20° to 40° Sun | Lighting handled only by planning |

| # | Source | What it did | Key finding | How we use it |
|---|---|---|---|---|
| 26 | Cauligi et al. (2023), IEEE Aerospace (ShadowNav) | Crater localisation at night or in shadow | Adds a light and stereo camera; under 10 m error in simulation | Hardware fix for shadow |
| 27 | Candan and Servadio (2026), arXiv | Deep-learning crater detector with an EKF | Fixed camera noise; recovers from a 5 km start error to a few hundred metres | Camera noise ignores lighting |
| 28 | Cushen et al. (2024–25), ARC-LIGHT | Built a 0.91 x 0.91 x 1.2 m dust chamber with camera, lidar and mist; trained a small network to estimate dust thickness from images | Uses the dust estimate to correct lidar distances and shuts itself off if error is too large; won NASA's 2024 Human Lander Challenge | Same trust idea for plume dust; future extension |

# References

Each item was checked against the linked page. *Notes* flag corrections to the original project brief or items still to confirm.

**Missions**


1. ispace (2023). HAKUTO-R Mission 1 debriefing (CTO R. Ujiie), PDF; press release of 26 May 2023. *Note:* the lander did not hover at 5 km; it believed it was at 0 km, descended slowly until propellant ran out, then free-fell.
2. SpacePolicyOnline (2025). Hardware failure doomed ispace's Mission 2.
3. SpacePolicyOnline (2025). IM-2's imperfect landing due to altimeter interference, South Pole lighting conditions; Spaceflight Now. Source is the Q1 2025 earnings call.
4. New Atlas (2024), Odysseus broke leg on landing; The Space Review (2024), article 4765. *Note:* IM-1's cause was a laser safety switch left on, not slopes or shadows.
5. Mehta, J. (2023). Chandrayaan-3 makes historic touchdown on the Moon. *Scientific American*, 23 Aug 2023.
6. JAXA (2024). SLIM landing results press briefing, 25 Jan 2024.
7. Freeman, R. H. (2025). Soft landing on the Moon: the ups and downs of robotic lunar landers. *Journal of Space Operations & Communicator* 21(2). Qualitative review of 12 news reports; background only.

### NASA priorities and architectures


8. NASA STMD (2026). FY26 Civil Space Shortfall Prioritization; 2026 Civil Space Shortfalls. *Note:* "ESDMD #1101" is a cross-reference inside need statement 3.02, not a ranking; 3.02 is one of 40 FY26 focus areas, not a "Top-5 agency priority".
9. Sostaric, R. (project manager), Carson, J. M. III (PI). SPLICE project poster, NASA. *Note:* a specific "Sostaric et al. (2021)" paper was not found; confirm on NTRS before citing it.
10. Getchius, J. et al. (2022). Hazard Detection and Avoidance for the Nova-C Lander. Venue not confirmed.
11. Lidar-based TRN for precise lunar landing, NASA (CORE).


### Terrain and illumination data


12. Barker, M. K., Mazarico, E., Neumann, G. A., Smith, D. E., Zuber, M. T., Head, J. W. (2021). Improved LOLA elevation maps for South Pole landing sites: error estimates and their impact on illumination conditions. *Planetary and Space Science* 203, 105119. doi:10.1016/j.pss.2020.105119.
13. Barker, M. K. High-Resolution LOLA Topography for Lunar South Pole Sites, NASA GSFC PGDA Product 78.
14. Mazarico, E., Neumann, G. A., Smith, D. E., Zuber, M. T., Torrence, M. H. (2011). Illumination conditions of the lunar polar regions using LOLA topography. *Icarus* 211(2), 1066–1081.
15. Bertone, S., McKenna, T. E., Barker, M. K., Mazarico, E., Beyer, R. A., Petro, N. Enhanced topography models for selected lunar South Pole regions with Shape-from-Shading. *The Planetary Science Journal*, doi:10.3847/PSJ/ae5b70 (PGDA 104). Confirm the year from the DOI page.


### Filtering theory


16. Bar-Shalom, Y., Li, X. R., Kirubarajan, T. (2001). *Estimation with Applications to Tracking and Navigation*. Wiley.
17. Mehra, R. K. (1972). Approaches to adaptive filtering. *IEEE Transactions on Automatic Control* 17(5), 693–698.
18. Hostetler, L. D., Andreas, R. D. (1983). Nonlinear Kalman filtering techniques for terrain-aided navigation. *IEEE Transactions on Automatic Control* 28(3), 315–323.


### Closest prior work


19. Adams, D. W., Ward, K. C., Smith, K. W., Pedrotty, S. M., Mendeck, G. F. (2025). Topographic slant range modeling and fault detection for precision planetary landing. AIAA SciTech Forum, Orlando. NTRS 20240014100.

20. Karlgaard, C. D., Lugo, R. A., Shidner, J. D., Hines, G. D., Amzajerdian, F., Gragossian, A., Litton, D. K., Angster, S. R. (2025). Navigation Doppler Lidar performance assessment and trajectory reconstruction of the Intuitive Machines IM-1 lunar landing. AAS/AIAA Astrodynamics Specialist Conference; NASA TM, NTRS 20250010290.
21. Prakash D., S. R., Raha, S. (2026). Co-state based data fusion and risk aware filtering for spacecraft navigation and hazard prediction. arXiv:2604.20485 (preprint).
22. Sheppard, A., Reale, C., Skinner, K. A. (2025). A stochastic approach to terrain maps for safe lunar landing. arXiv:2512.12058; IEEE Aerospace Conference 2026.
23. Cowan, S., Fanti, P., Williams, L. B. S., Yam, C. H., Asakuma, K., Nada, Y., Izzo, D. (2025). Vision-guided optic flow navigation for small lunar missions. arXiv:2511.17720.
24. Tomasicchio, G., Andolfi, L., Brancati, M., Di Donna, A. M., Giannattasio, S., Del Prete, R., Ostrogovich, L., Renga, A., Grassi, M., Ceresoli, M., Silvestrini, S., Lavagna, M. (2024). ION GNSS+ 2024; summarised in *Inside GNSS*, 28 Apr 2025. Confirm the exact title in the ION proceedings.
25. Shoemaker, M. A., et al. (2022). Terrain relative navigation in a lunar landing scenario using autoNGC. AIAA SciTech 2022. NTRS 20210024481.
26. Cauligi, A., Swan, R. M., Ono, M., Daftry, S., Elliott, J., Matthies, L., Atha, D. (2023). ShadowNav: crater-based localization for nighttime and permanently shadowed region lunar navigation. IEEE Aerospace Conference 2023.
27. Candan, B., Servadio, S. (2026). Deep learning-based lunar crater terrain relative navigation. arXiv:2606.14776 (preprint).
28. Cushen, A. et al. ARC-LIGHT, NASA Human Lander Challenge 2024 (U-M CLaSP); *Aerospace* 12(3), 177, 2025 (MDPI). HuLC technical paper team: A. Cushen, S. Carrico, A. Bueno, C. Wettstein, J. I. Adalja, M. Shi, N. Garcia, Y. Garcia; advisors M. Gamba and C. Ruf. Check the MDPI author list before citing as "Cushen et al. (2025)".

**Added**

29. Button, K. (2025). How Intuitive Machines is planning to make its third Moon landing a success. *Aerospace America* (AIAA), 22 Jul 2025.